



Operating system and system programming (25CS2104E)

Linux System Information and Resource Monitoring Tool

Review-I

Under the Esteemed Guidance of

Raghupathi M
CSE Department

Team Members:

2520090090 - Ch. Sai Vihamsh

2520090016 - P. Rahul

2520090008 - B. Sathish Kumar

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Abstract

- Linux-based tool for monitoring system resources.
- Monitors CPU, memory, disk, system uptime, and running processes.
- Collects information using Linux commands and `/proc` file system.
- Displays information through a simple Command Line Interface (CLI).
- Identifies processes using high CPU or memory.
- Demonstrates OS concepts like process management, memory management, file systems, system calls, and resource utilization.

Introduction

- Linux continuously manages resources such as CPU, RAM, disk, and processes.
- Users may find it difficult to understand how these resources are being utilized.
- Our project provides a simple tool to collect and display system information.
- It helps users observe the current system status and identify resource-intensive processes.
- The project demonstrates practical concepts of Operating Systems and System Programming.

Software requirements

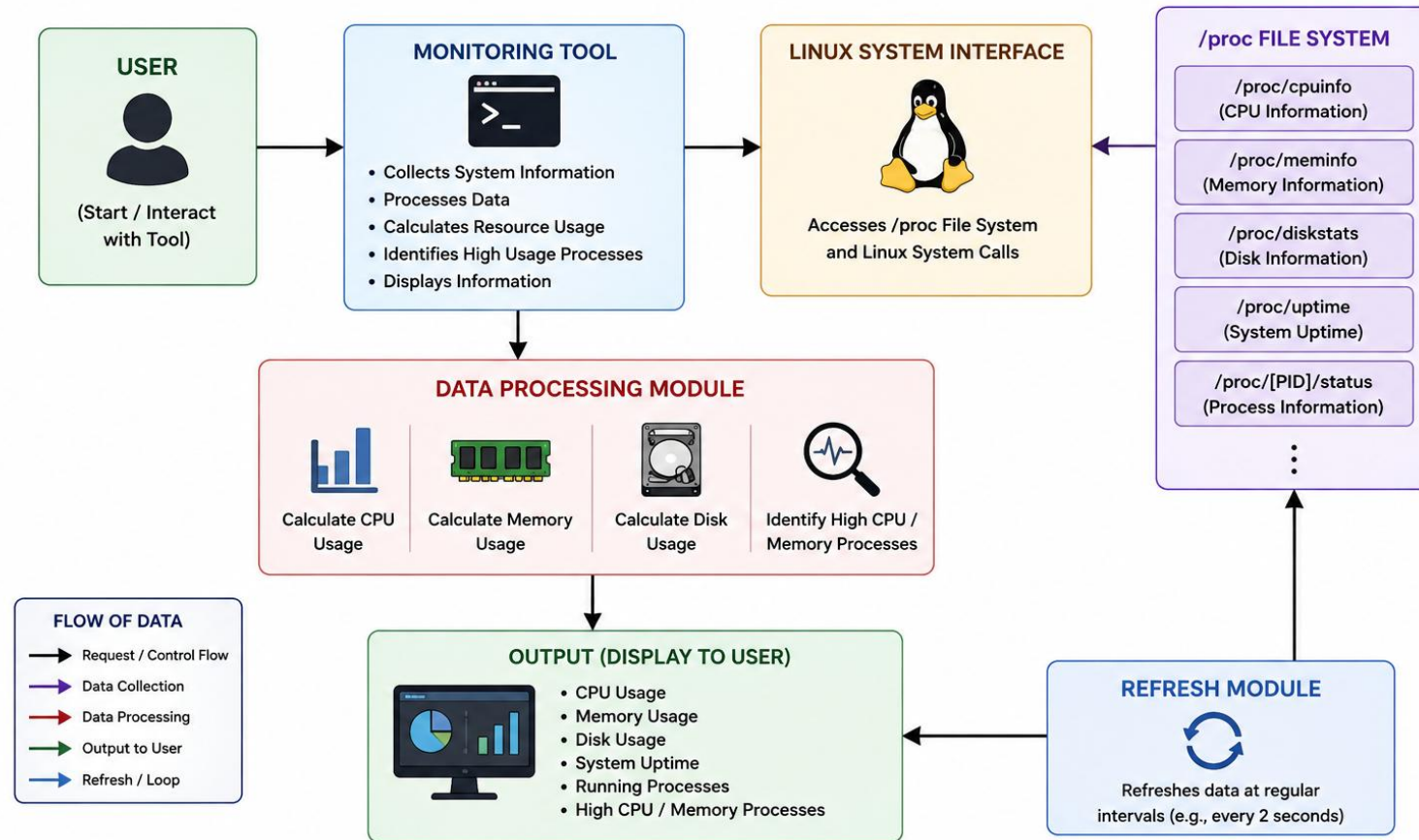
Software	Specification
Operating System	Linux / Ubuntu
Programming Language	C / C++
Compiler	GCC
System Interface	Linux /proc File System
Interface	Command Line Interface (CLI)

Hardware requirements

Hardware	Specification
Processor	Intel / AMD Processor
RAM	Minimum 4 GB
Storage	Minimum 20 GB
Computer	Laptop / Desktop

System Architecture

SYSTEM ARCHITECTURE



Conclusion

- The project provides a simple Linux system monitoring solution.
- It monitors CPU, memory, disk, uptime, and running processes.
- It helps identify processes consuming high system resources.
- The project provides practical understanding of Linux and Operating System concepts.
- It demonstrates process management, memory management, file systems, /proc, system calls, and resource utilization.
- The final tool will be lightweight, simple, and easy to understand.

Thank you